

Maximum Sum Subarray

of Size K



k = 3

2	6	43	4	5
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Max sum: **0**

Window sum: **0**

k = 3

2	6	43	4	5
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Max sum: **0**

Window sum: **2**

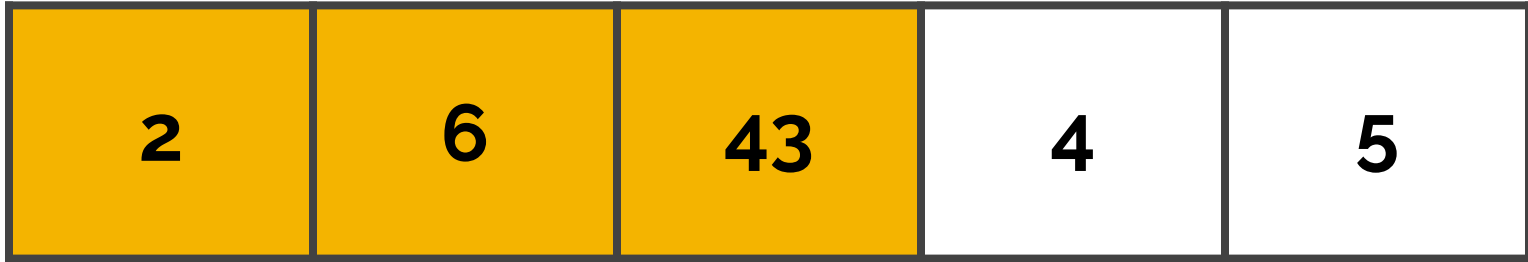
k = 3



Max sum: **0**

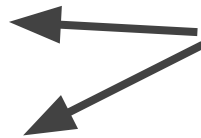
Window sum: **8**

$k = 3$



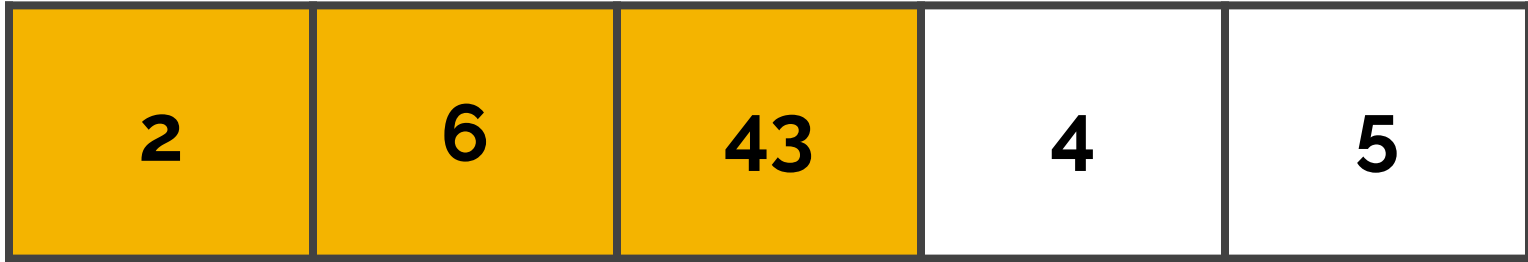
Max sum: **0**

Window sum: **51**



Compare

k = 3



Max sum: **51**

Window sum: **51 - 2 = 49**



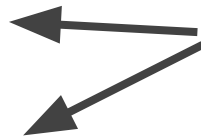
*Subtract
window start*

k = 3



Max sum: **51**

Window sum: **53**



Compare

k = 3



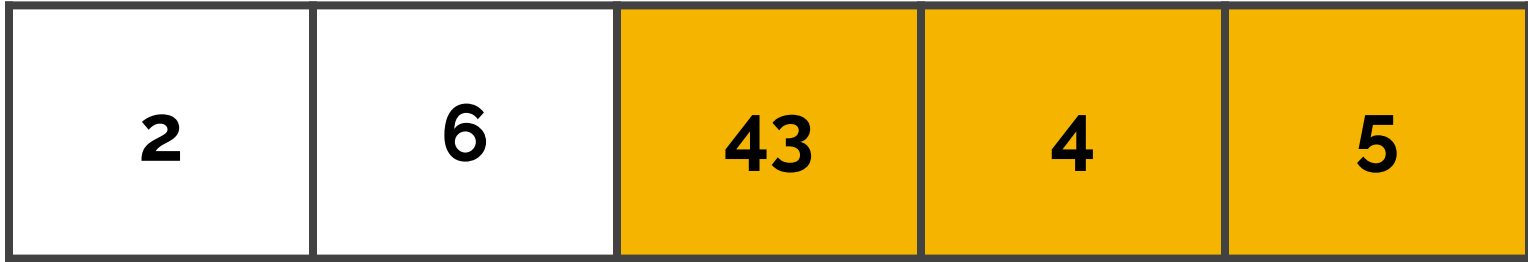
Max sum: **53**

Window sum: **53 - 6 = 47**



*Subtract
window start*

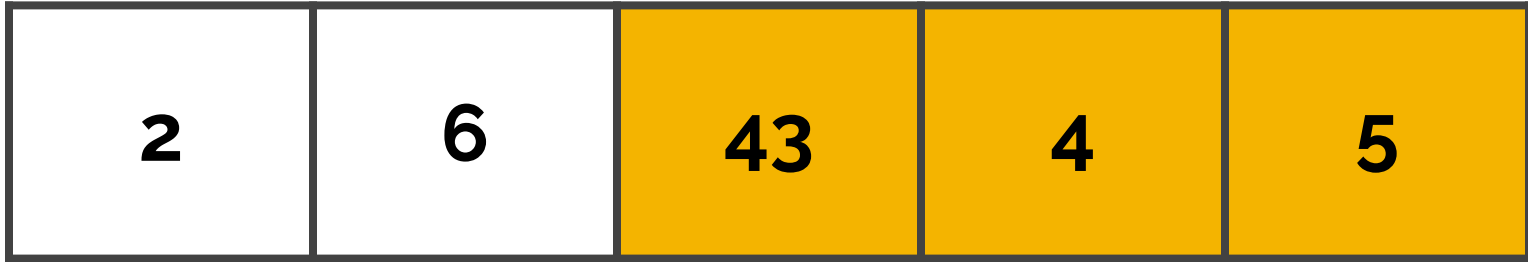
k = 3



Max sum: **53**

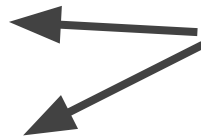
Window sum: **47 + 5 = 52**

$k = 3$



Max sum: **53**

Window sum: **52**



Compare

Time complexity **$O(N)$**

Space complexity **$O(1)$**